

Problem

Use the convolution integral to find:

(a) $t * e^{at}u(t)$

(b) $\cos(t) * \cos(t)u(t)$

Step-by-step solution

Step 1 of 6

(a)

Consider the following functions:

$$x(t) = t \text{ and } h(t) = e^{at}u(t).$$

Step 2 of 6

Write the convolution integral of a system to an excitation $x(t)$, knowing the system impulse response $h(t)$.

$$y(t) = x(t) * h(t)$$

$$y(t) = \int_0^t x(\lambda)h(t-\lambda)d\lambda$$

For the functions $x(t) = t$ and $h(t) = e^{at}u(t)$ the above expression becomes:

$$\begin{aligned} y(t) &= \int_0^t \lambda e^{a(t-\lambda)}d\lambda \\ &= e^{at} \int_0^t \lambda e^{-a\lambda}d\lambda \\ &= e^{at} \left[-\frac{e^{-a\lambda}}{a^2}(a\lambda + 1) \right]_0^t \\ &= e^{at} \left[-\frac{e^{-a(t)}}{a^2}(a(t) + 1) - \left(-\frac{e^{-a(0)}}{a^2}(a(0) + 1) \right) \right] \end{aligned}$$

Simplified further...

$$\begin{aligned} y(t) &= e^{at} \left[-\frac{e^{-at}}{a^2}(at + 1) - \left(-\frac{1}{a^2} \right) \right] \\ &= e^{at} \left[-\frac{e^{-at}}{a^2}(at + 1) + \frac{1}{a^2} \right] \\ &= \frac{e^{at}}{a^2} - \frac{1}{a^2}(at + 1) \end{aligned}$$

Step 3 of 6

Write the expression in desired form.

$$y(t) = \frac{t}{a}(e^{at} - 1) - \frac{1}{a^2} - \frac{e^{at}}{a^2}(at - 1) \quad \text{since} \left\{ \begin{array}{l} \frac{e^{at}}{a^2} - \frac{1}{a^2}(at + 1) = \\ \frac{t}{a}(e^{at} - 1) - \frac{1}{a^2} - \frac{e^{at}}{a^2}(at - 1) \end{array} \right\}$$

Therefore the convolution of $x(t) * h(t)$ is $\boxed{\left(\frac{t}{a}(e^{at} - 1) - \frac{1}{a^2} - \frac{e^{at}}{a^2}(at - 1) \right) u(t)}$.

Step 4 of 6

(b)

Consider the following functions:

$$x(t) = \cos(t) \quad \text{and} \quad h(t) = \cos(t)u(t).$$

Step 5 of 6

Write the convolution integral of a system to an excitation $x(t)$, knowing the system impulse response $h(t)$.

$$y(t) = x(t) * h(t)$$

$$y(t) = \int_0^t x(\lambda)h(t - \lambda)d\lambda$$

For the functions $x(t) = \cos(t)$ and $h(t) = \cos(t)u(t)$ the above expression becomes:

$$\begin{aligned} y(t) &= \int_0^t \cos(\lambda)\cos(t - \lambda)d\lambda \\ &= \int_0^t \cos(\lambda) \left(\cos(t)\cos(\lambda) + \sin(t)\sin(\lambda) \right) d\lambda \quad \text{since} \left\{ \begin{array}{l} \cos(A - B) = \\ \cos A \cos B + \sin A \sin B \end{array} \right\} \\ &= \int_0^t (\cos t \cos^2 \lambda + \sin t \cos \lambda \sin \lambda) d\lambda \end{aligned}$$

Step 6 of 6

Simplify further...

$$\begin{aligned}
 y(t) &= \cos t \int_0^t \cos^2 \lambda d\lambda + \sin t \int_0^t \sin \lambda \cos \lambda d\lambda \\
 &= \cos t \int_0^t \left(\frac{1 + \cos 2\lambda}{2} \right) d\lambda - \sin t \int_0^t \cos \lambda d(\cos \lambda) \quad \text{since } \left\{ \begin{array}{l} \frac{d}{dt}(\cos t) = \\ -\sin t \end{array} \right\}
 \end{aligned}$$

Assume $x = \cos \lambda$ in the second integral term,

Then limits change from $(0, t)$ to $(1, \cos t)$.

$$\begin{aligned}
 y(t) &= \cos t \int_0^t \left(\frac{1 + \cos 2\lambda}{2} \right) d\lambda - \sin t \int_1^{\cos t} x dx \\
 &= \frac{\cos t}{2} \int_0^t d\lambda + \frac{\cos t}{2} \int_0^t \cos 2\lambda d\lambda - \sin t \left[\frac{x^2}{2} \right]_1^{\cos(t)}
 \end{aligned}$$

Simplified further:

$$\begin{aligned}
 y(t) &= \frac{\cos t}{2} [\lambda]_0^t + \frac{\cos t}{2} \left[\frac{1}{2} \sin 2\lambda \right]_0^t - \frac{\sin t}{2} [(\cos t)^2 - 1] \\
 &= \frac{t \cos t}{2} + \frac{\cos t}{2} \left[\frac{1}{2} \sin 2t - \frac{1}{2} \sin 2(0) \right] - \frac{\sin t}{2} [(\cos^2 t) - 1] \\
 &= \frac{t \cos t}{2} + \frac{\cos t}{2} \left[\frac{1}{2} \sin 2t - 0 \right] - \frac{\sin t}{2} [\cos^2 t - 1] \\
 &= 0.5 \cos(t) (t + 0.5 \sin 2t) - 0.5 \sin(t) [\cos^2(t) - 1]
 \end{aligned}$$

Therefore the convolution of $x(t) * h(t)$ is

$$\boxed{[0.5 \cos(t) (t + 0.5 \sin(2t)) - 0.5 \sin(t) (\cos^2(t) - 1)] u(t)} .$$